

Ansh Jetli

Central NJ (NYC Metro) | github.com/a-jetli | linkedin.com/in/anshjetli | ajetli.com | ansh.jetli@gmail.com

EDUCATION

Rutgers University - New Brunswick

New Brunswick, NJ

Bachelor of Science in Computer Science · GPA: 3.4

Graduating Dec 2026

Relevant Coursework: *Data Structures, Computer Architecture & Assembly, Design and Analysis of Computer Algorithms, Principles of Programming Languages, Software Methodology, Systems Programming, Intro to AI*

PROJECTS

RAG Pipeline over AI/ML Papers | *Python, LangGraph, FastAPI, ChromaDB, OpenAI, Pydantic* May 2026

- Built hybrid retrieval over 1,000 arXiv papers by combining OpenAI embedding search with BM25 through Reciprocal Rank Fusion, then reranking a 24-candidate pool with a small local model.
- Built a 5-node LangGraph workflow that decomposes questions into focused searches, grades retrieved context for sufficiency, performs at most one reformulation retry, and accumulates deduplicated context across passes.
- Improved compound-query execution by batching all sub-query embeddings into one OpenAI request and running each retrieval and reranking path concurrently.
- Deployed as Dockerized FastAPI service on Hugging Face Spaces, loading a prebuilt BM25 index and persistent ChromaDB collection during startup instead of rebuilding the keyword index on the deployed path.
- Reduced median cross-encoder local rerank latency 95% (1.28s to 63ms across 120 calls) by replacing FlashRank MiniLM with TinyBERT.

Rotation - Last.fm Analytics Tool | *Python, FastAPI, PostgreSQL, Docker Compose, Amazon EC2* July 2026

- Engineered FastAPI/PostgreSQL history and analytics pipeline that imports complete Last.fm listening history and incrementally syncs new data using ON CONFLICT checks and advisory locks to make overlapping jobs retry safe.
- Implemented 16 analytics endpoints using CTEs and window functions for insights like streak and binge detection instead of processing full histories in Python.
- Built a content based recommender pulling user-defined Last.fm genre tags through a curated blocklist, TF-IDF, cosine similarity, all precomputed and cached per user periodically with a 90 day recency bias.
- Containerized and deployed to AWS EC2 instance with Docker Compose; wrote test suite to run against disposable PostgreSQL instance that surfaced a timezone bug in 7 of 10 date-bucketing queries; configured GitHub Actions to run test suite on every push, blocking deploys that fail.

LLM-Driven Text RPG Engine | *Python, OpenAI API, Pydantic, Textual UI* May 2026

- Built an open ended, genre agnostic text RPG in Python and Textual TUI with abstract game mechanics and an LLM narrator, allowing total narrative freedom while the engine retains authoritative control over game state.
- Defined Pydantic based OpenAI structured output for seamless LLM to engine interfacing; the schema enforces response structure and types while the engine enforces game rules; the model suggests, but the engine enforces.
- Designed a three-tiered context pipeline for unbounded play within fixed token window - verbatim recent turns, a rolling summary, and completed arcs frozen as cold context volumes so summarization doesn't erode early details later in the story; authoritative state persisted as JSON, with region and NPC prose logged to Markdown on disk.
- Built retrieval engine that pulls location-relevant region/NPC/facts as length capped entries, while a regex turn classifier drives a director to inject matching procedure step and rolls eligible weighted random events into each injected prompt.

TECHNICAL SKILLS

Languages: Python, Java, SQL, OCaml

AI/ML: LangGraph, RAG, OpenAI API, Pydantic structured outputs, embeddings, cross-encoder reranking, BM25, prompt evaluation

Backend and Data: FastAPI, REST APIs, ChromaDB, AWS, Docker

Tools: Git, GitHub, VS Code, Ollama (local LLMs), LaTeX